Introduction to Computer Science 2.0

9th September 2024

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

SECTION A - Multiple Choice Question
Circle ONE (1) correct answer from A, B, C, or D for each question.
Each question is worth 1 mark.

Question 1

Which ONE (1) of the following is **not** internal hardware?

A CPU

B Graphics card

C Keyboard

D Hard drive

1 mark

Question 2

Which of these is an example of a phishing attack?

A Downloading software containing a virus

B Attempting to gain sensitive information by impersonating a business

C Attempting to overload a server to cause it to crash

D Attempting to decode an encrypted message

1 mark

Question 3

What is the primary function of application software?

A To manage computer hardware

B To perform specific tasks for users

C To connect computers over a network

D To cool down the computer components

1 mark

Question 4

Which component of the computer architecture is responsible for temporarily storing frequently accessed data?

A Cache

B CPU

C ROM

D Hard Drive

1 mark

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Mark

Question 5

If P = 1 and Q = 1, what is NOT (P AND Q)

A 0

B 1

C -1

D 2

1 mark

Question 6

What is the purpose of input devices in a computer system?

A To provide data and instructions to the computer

B To display the output produced by the computer

C To perform calculations and logical operations

D To store large amounts of data permanently

1 mark

Question 7

Which of these is likely to have a more stable connection?

A A wireless PAN using Wi-Fi

B A wired LAN using Ethernet cables

C 4 handheld devices connected by Bluetooth

D A wireless LAN using Wi-Fi

1 mark

Question 8

A multi-user network that is in the same building and is limited in distance it covers, is called a…

A PAN

B WAN

C MAN

D LAN

1 mark

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Question 9

A large amount of e-waste is created as a result of people…

A Frequently using cars to drive to work

B Putting devices into sleep mode

C Frequently upgrading their electronic device

D Not recycling their food waste

1 mark

Question 10

What is the purpose of secondary storage in a computer system?

A To store data permanently

B To temporarily hold data for processing

C To increase the speed of the CPU

D To manage network connections

1 mark

Question 11

Which of these passwords is the strongest?

A computerscience

B C0mputerSci3nc3

C C0mputer?Sci3nce3!

D Computer_Science

1 mark

Question 12

Comparing countries that have access to the latest computer technology and those that do not, is called….

A The Computing Digitalisation

B The Global Digitalisation

C The Computing Divide

D The Global Divide

1 mark

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Question 13

If the letter ‘q’ has the ASCII value 113, what would the ASCII value 116 represent?

A **r**

B u

C t

D s

1 mark

Question 14

Which is an advantage of storing data in the Cloud?

A It is more secure

B It can be accessed from anywhere with an Internet connection

C You can hold multiple copies

D It is faster to retrieve

1 mark

Question 15

Which of these is a reason someone may chose magnetic tape as a form of storage?

A Lightweight

B Easy to read/write from

C Low Cost

D High Storage Capacity

1 mark

Question 16

Which of these is a valid example of hexadecimal representation of a colour?

A #BBA4

B #34AA22BB67

C #BBAACC

D #BLUE56

1 mark

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Mark

Question 17

Repetitive Strain Injury (RSI) is caused by

A Workers using computers for a long period without a break

B Sun glare on the monitor

C Unsafe cables and wiring

D Lifting heavy machinery

1 mark

Question 18

What is “multitasking” in the context of an operating system?

A The ability to upgrade hardware components

B The ability to execute several programs at the same time

C The process of deleting old files

D The process of installing new software

1 mark

Question 19

What is the main difference between 3G, 4G and 5G technology?

A The size of the devices

B Processor speeds

C Data transfer rate

D Materials used in production

1 mark

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Question 20

What is run length encoding?

A An example of lossless compression

B An example of lossy compression

C An example of a computer virus

D A programming language

1 mark

Question 21

What is a potential **advantage** of using a compact disc (CD) as a form of secondary storage?

A Data transfer speed

B Durability

C High storage capacity

D Lightweight

1 mark

Question 22

What is bandwidth?

A The amount of data that can be transferred in a given time

B A specific frequency band using by Wi-Fi

C A measure used in Smart TV's

D The amount of pixels a photo stores

1 mark

Question 23

Which is an advantage of a colour ink-jet printer over a colour laser printer?

A Is less expensive upfront cost

B Is faster

C Has a better-quality print

D Is portable

1 mark

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Mark

Question 24

What is Internet Censorship?

A The act of a boss monitoring emails and websites of employees

B Removing access to the internet in places of education

C Controlling what others can access and publish on the Internet

D A filtering website

1 mark

Question 25

What role does the ALU play during the Fetch, Decode, Execute cycle?

A It controls the flow of data between the CPU and memory

B It stores instructions and data temporarily

C It performs arithmetic and logical operations

D It manages the output devices

1 mark

Question 26

Which of these is **NOT** an example of an embedded system?

A Gaming Console

B Washing Machine

C Smartphone

D Network Server

1 mark

Question 27

Which of the following is **NOT** an example of application software?

A Microsoft Word

B Adobe Photoshop

C Windows 10

D Google Chrome

1 mark

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Question 28

Firewire and SCSI are examples of:

A Buses

B Connectors

C Processors

D Anti-malware software

1 mark

Question 29

Which of these examines data entering a network in order to block any potential threats?

A Bridge

B Router

C Firewall

D Switch

1 mark

Question 30

Which internal component is often measured in gigabytes?

A CPU

B RAM

C Clock

D Cache

1 mark

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Mark

SECTION B
ANSWER ALL QUESTIONS

Question 31

a) NCC Book Shop has installed a new computer system for recording the sales of books.

i) Using examples of a book shop, describe the difference between data and information. **3**

ii) A customer wants to check if a book called “The Great Gatsby” is in stock. **3**

Describe the input, process and output that would take place in the computer system for this type of query.

iii) Describe TWO (2) input devices that would be needed for a member of staff to carry out a check for customers. **2**

iv) A large digital display screen shows customers videos promoting new books. **1**

State the most suitable type of connection for the screen to the computer system.

v) The previous computer system used had a command line interface. **1**

State ONE (1) feature of a command line interface.

Total 10 Marks

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Mark

Question 32

a) Match the CPU registers to their purpose. State the name of the missing register labelled (3). **5**

1. MDR	a) Holds the memory address of the next instruction to be fetched from primary memory
2. Program Counter	b) Holds the data being processed and the results of processing
3. _____	c) Holds the instruction that is currently being decoded and executed
4. MAR	d) Holds the address of the current instruction or data that is to be fetched from memory
5. CIR	e) Holds the contents found at the address held in the MAR

b) State THREE (3) internal hardware factors that affect a CPU's performance. **3**

c) Software is created using a programming language.

i) Machine code is often referred to as a 'First Generation' Programming Language. **1**

State what categorises a language to be classified as a 'First Generation' language.

ii) State the generation of languages that contains Java, Python and C++. **1**

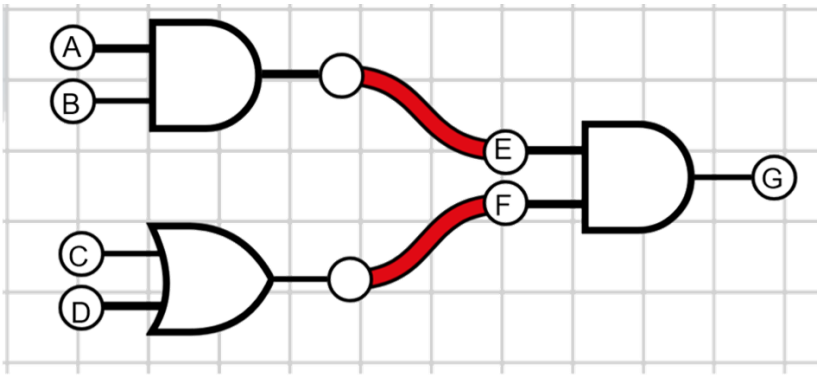
Total 10 marks

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Mark

Question 33

a) A logic circuit has been created as shown below: **7**



Input A is always fixed as the value 1 or TRUE.

Create a truth table that shows all other possible inputs (B-F) and output (G).

b) Run length encoding (RLE) is one method of compressing data.

i) Give TWO (2) reasons why data compression is often used. **2**

ii) State the feature of data that allows it to be compressed efficiently using RLE. **1**

Total 10 marks

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Mark

Question 34

a) A TV production is recording bird calls for a nature programme. **3**

Explain how the sound waves they collect are converted so they can be stored in a computer for editing.

b) A student has recorded a 30 second digital soundtrack using a sample rate of 1000Hz. **3**

8 bits have been used to store each sample taken.

Calculate the file size in kilobytes of the digital soundtrack.

You should show your working.

c) State the binary **and** denary representations of the following hexadecimal codes:

i) A1 **2**

ii) ED **2**

Total 10 Marks

Question 35

a) Servers and MAC addressing are both used in sending data across a computer network.

i) What is a computer network? **1**

ii) Explain the purpose of a proxy server. **2**

iii) Explain the purpose of a MAC address **and** give ONE (1) example of what a MAC address looks like. **2**

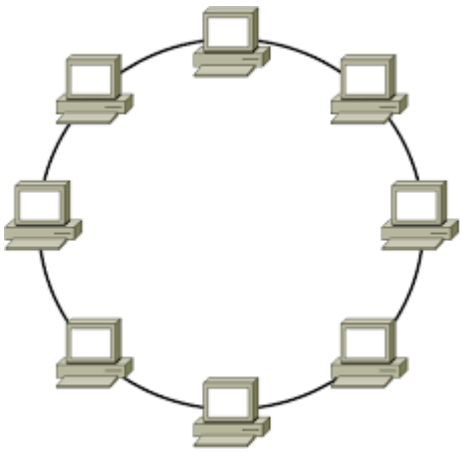
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Mark

b) An airport is installing a wireless network. **4**

Discuss TWO (2) problems airport staff and customers may experience with a wireless network.

c) State the name of this network topology **1**



Total 10 Marks

Question 36

a) TWO (2) students are discussing their views of the Internet and Security.

Student A stated

"The World Wide Web and the Internet are essentially the same thing"

Student B stated:

"Untrained employees are not really that much of a threat to a company's network security"

i) Explain why Student A is incorrect. **2**

ii) Explain, with examples, why Student B is incorrect. **2**

iii) What is the purpose of a network policy? **1**

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Mark

b) Penetration Testing and Biometrics are terms connected to network security

i) Define the term 'penetration testing'. **1**

ii) Describe the difference between black-box and white-box penetration testing. **2**

iii) Explain **TWO (2)** ways in which biometrics could be used to prevent a hacker gaining access to highly confidential files. **2**

Total 10 Marks

Question 37

a) "Mobile phones should be completely banned in schools and places of education." **8**

Compare arguments for and against this viewpoint. You should discuss:

- Ethical impacts
- Social impacts
- Environmental impacts
- Health and Safety

b) Automated machines are replacing humans in many jobs. **2**

State **ONE (1)** job where it would currently be difficult for humans to be replaced by machines and explain why.

Total 10 marks

End of paper

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